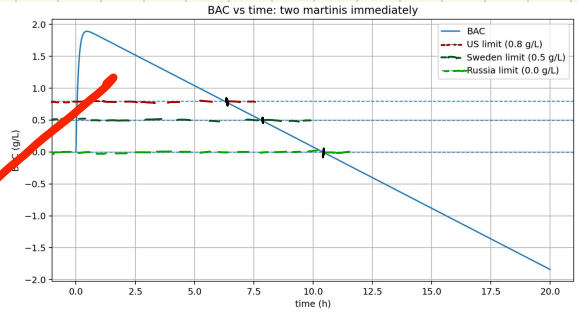


Problem Set 6

8-5

i) Same time

- a) 6.28 h
- b) 7.813 h
- c) 10.417



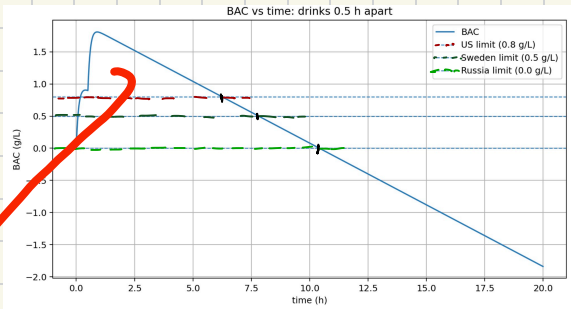
h) Lower Body Fluid Volume would mean that the person would have to wait longer before they could drive.

higher body would be opposite the ethanol would process faster.

i) Don't drink and drive.

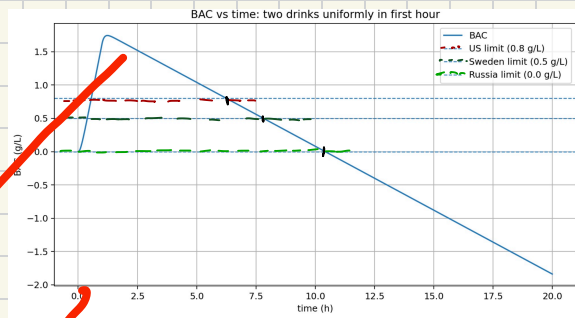
d) $\frac{1}{2}$ hr apart

US = 6.25 hr
Sweden = 7.813 hr
Russia = 10.417

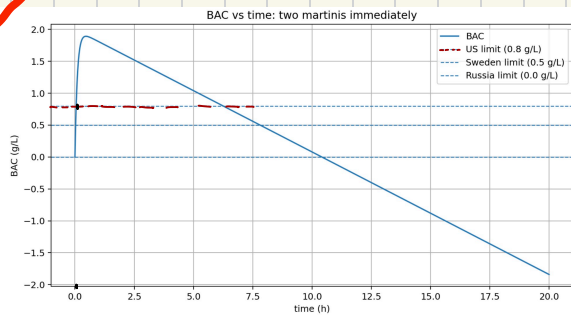


e) Uniformly

US - 6.274 hr
Swe - 7.836 hr
Rus - 10.441 hr

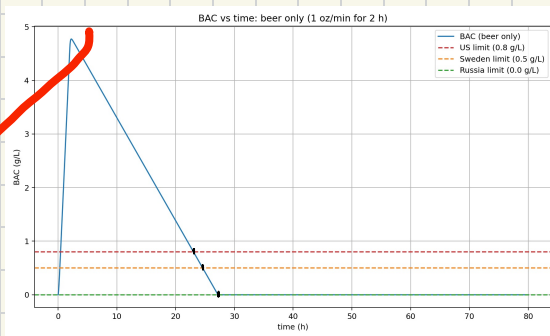


f) about - 3.12 hrs



g) Beer for 2 hrs

US - 23.05 hrs
Swe - 24.614 hrs
Rus - 27.22 hrs



8-7

A)

$$\frac{k_0}{k_u} = \frac{A_0}{A_u} \exp\left(-\frac{E_0 - E_u}{RT}\right) = \frac{10}{100} \exp\left(-\frac{(-8000) - (-1000)}{T}\right)$$

$$S = \frac{k_0}{k_u} C_A^\alpha C_B^\beta$$

$$\begin{array}{ll} E_0 > E_u & \text{high } T \text{ s} \\ \alpha_1 > \alpha_2 & \text{high } C_A \text{ s} \\ \beta_1 > \beta_2 & \text{low } C_B \text{ s} \end{array}$$

Semi batch

Would be good in order to add B slowly to A at high T.

- b) Either PFR or CSTR would work, same concentration are the same, keep at low T to maximize desired product.
Undesired has high E_a - so low T $\rightarrow \downarrow E_a$
- c) Use PFR - parallel reaction, in PFR, we can lower have lower concentration of B so less of it reacts with our desired product. Run at low T as undesired product has high E_a .
- d) Use PFR - At low T, we can continuously remove D to prevent the 2nd reaction. Both unwanted reactions have higher E_a than desired product.
 $\rightarrow$ Also low C_D towards end.
- e) Use PFR as this will get optimal D product, As reaction continues A will drop which leads to more D. High T as E_a is higher compared to other.
- f) Semi-batch - high T - B goes slowly so low C_B , and D can be removed as it's formed.
Low C_B will prevent $B + D \rightarrow U_2$.

8-11

$$1) \quad r_{1A} = -7_{\text{mm}^{-1}} \cdot .1 \frac{\text{mol}}{\text{dm}^3} = -.7 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_{2D} = 3 \frac{\text{dm}^6}{\text{mol}^2 \text{min}} \left(0.51 \frac{\text{mol}}{\text{dm}^3}\right)^2 \left(0.1 \frac{\text{mol}}{\text{dm}^3}\right) = 0.078 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_{2A} = -\frac{1}{3} r_{2D} = -0.026 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_{3A} = 0 \frac{\text{mol}}{\text{dm}^3 \text{min}}, \quad \text{not in rxn}$$

$$2) \quad r_{1B} = \frac{1}{3} r_{1A} = 0.233 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_{2B} = r_{3B} = 0, \quad \text{OB in rxn}$$

$$3) \quad r_{1C} = \frac{1}{3} r_{1A} = 0.233 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_{2C} = -\frac{2}{3} r_{2D} = -0.052 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_{2C} = r_{2E} \quad r_{2C} = -k_E C_D C_C = -2 \frac{\text{dm}^3}{\text{mol min}} \left(0.049 \frac{\text{mol}}{\text{dm}^3}\right) \left(0.51 \frac{\text{mol}}{\text{dm}^3}\right) = -0.05 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$4) \quad r_{1D} = 0$$

$$r_{2D} = 0.078 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_{3D} = \frac{1}{3} r_{3E} = -0.0667 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$5) \quad r_{1E} = r_{2E} = 0$$

$$r_{3E} = k_E C_D C_C = 0.05 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$6) \quad r_A = \sum r_A = -.7 + -.026 = -.726 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_B = 0.233 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_C = 0.233 - 0.052 - 0.05 = .131 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_D = 0.078 - 0.0667 = 0.0113 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$r_E = 0.05 \frac{\text{mol}}{\text{dm}^3 \text{min}}$$

$$7) \quad V_0 = 100 \frac{\text{dm}^3}{\text{min}} \quad C_A = 3 \frac{\text{mol}}{\text{dm}^3}$$

$$C_A = C_{A0} \left(\frac{F_A}{F_{A0}} \right) \rho$$

$$\frac{d\rho}{dV} = -\frac{\alpha}{Z\rho} \frac{F_A}{F_{A0}}$$

$$F(C_A) = V_0 C_{A0} - V_0 C_A + r_A V = 0$$

$$\left(100\right) \frac{\text{dm}^3}{\text{min}} + \left(3\right) \frac{\text{mol}}{\text{dm}^3} - 100 \frac{\text{dm}^3}{\text{min}} \left(.1 \frac{\text{mol}}{\text{dm}^3}\right) =$$

$$.726 \frac{\text{mol}}{\text{dm}^3 \text{min}} V = 0$$

$$300 - 10 - .726 V = 0 = \frac{-290}{-.726} = 399.45 \text{ dm}^3$$

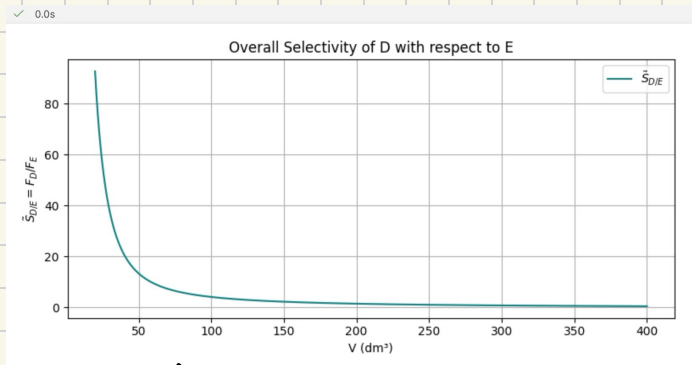
② $F_A = 1\% C_A = \frac{100 \text{ dm}^3}{\text{min}} \left(0.1 \frac{\text{mol}}{\text{dm}^3} \right) = 10 \frac{\text{mol}}{\text{min}}$

$F_S = \frac{100 \text{ dm}^3}{\text{min}} (0.93) = 93 \frac{\text{mol}}{\text{min}}$

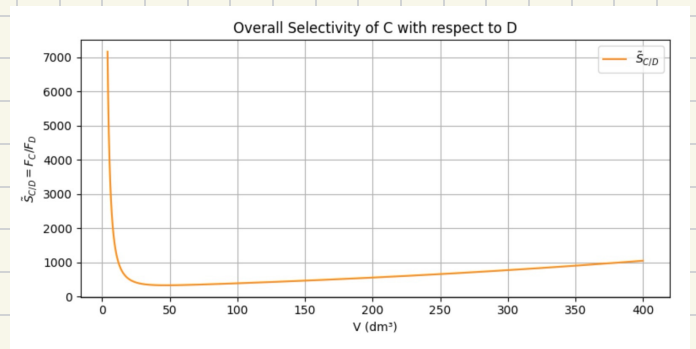
$F_C = 51 \frac{\text{mol}}{\text{min}}$

$F_D = 4.9 \frac{\text{mol}}{\text{min}}$

9)

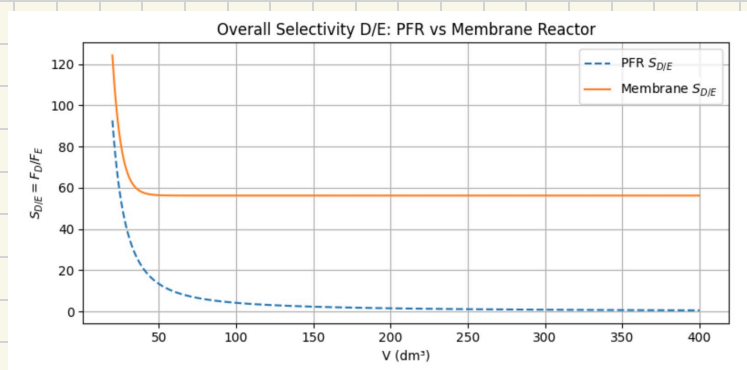
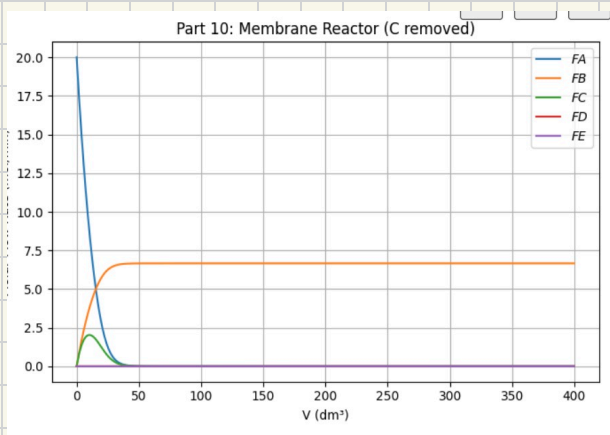


$D/E = 0.52$



$C/D = 1.03 \times 10^3$

10)



$S_{D/E} = 56.2717$

Membrane Reactor

